



Supplementary Materials for

Realization of fractional quantum Hall state with interacting photons

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DEVICE SETUP

Superconducting chip

The circuit QED lattice is assembled using flip-chip technology (Fig. S1). The top chip, referred to as the quantum chip, contains a photon box array and Transmon-based couplers that connect them. The photon boxes are based on Plasmonium qubits which is composed of two metallic Al islands (charging energy $E_C/h \sim 0.60$ GHz) connected with a superconducting inductance (inductive energy $E_L/h \sim 2.70$ GHz), and shunted by a nonlinear Josephson junction (Josephson energy $E_J/h \sim 4.58$ GHz). Neighboring photon boxes feature the tunable Hamiltonian (61)

$$\hat{H}/\hbar = \sum_{j=1,2} \frac{1}{2} \omega_j \hat{\sigma}_j^z + \frac{1}{2} \omega_c \hat{\sigma}_c^z + \sum_{j=1,2} J_j (\hat{\sigma}_j^+ \hat{\sigma}_c^- + \hat{\sigma}_j^- \hat{\sigma}_c^+) + J_{12} (\hat{\sigma}_1^+ \hat{\sigma}_2^- + \hat{\sigma}_1^- \hat{\sigma}_2^+), \quad (\text{S1})$$

where $\hat{\sigma}_\lambda^z$, $\hat{\sigma}_\lambda^+$, $\hat{\sigma}_\lambda^-$ ($\lambda = 1, 2, c$) are the Pauli-Z, raising and lowering operators of the qubits and coupler, respectively. In the dispersive regime where $J_j \ll |\omega_j - \omega_c|$, the coupler can be traced off, resulting in an effective tunable coupling

strength J_{eff} between photon boxes

$$\hat{H}/\hbar = \sum_{j=1,2} \frac{1}{2} \tilde{\omega}_j \hat{\sigma}_j^z + J_{\text{eff}} (\hat{\sigma}_1^+ \hat{\sigma}_2^- + \hat{\sigma}_1^- \hat{\sigma}_2^+), \quad (\text{S2})$$

where $\tilde{\omega}_j = \omega_j + \frac{J_j^2}{\omega_j - \omega_c}$ and $J_{\text{eff}} = \frac{1}{2} J_1 J_2 (\frac{1}{\omega_1 - \omega_c} + \frac{1}{\omega_2 - \omega_c}) + J_{12}$. Here, the couplers are in fact the Transmon qubits. They are designed to have SQUIDs with symmetric Josephson junctions which allow frequency tuning up to above 10 GHz. The Plasmoniums and couplers are coupled capacitively, with an estimated coupling strength of $J_j/2\pi \sim 250$ MHz, together with a direct coupling between neighboring qubits of $J_{12}/2\pi \sim 20$ MHz.

The bottom chip includes classical signal interfaces for control and readout. Here, each photon box is equipped with an individual line for frequency modulation and driving, each coupler is equipped with an individual line for frequency modulation. Each photon box is further coupled capacitively with a resonator for dispersive readout. The resonators are quarter-wavelength coplanar waveguides, which is capacitively coupled to a shared Purcell filter. We fabricated two Purcell filters in our sample, each responsible for the joint measurement of eight photon boxes with frequency multiplexing technique.

Our sample is based on the aluminum film on the low-loss sapphire substrate. The on-chip inductance is fabricated with a Josephson junction array, each consists 43 Al/AlO_x/Al Josephson junctions connected in series. On the bottom chip, we fabricated SiO₂/Al crossovers to alleviate slot-line modes and reduce signal crosstalk to adjacent qubits. On both of the top chip and the bottom chip, indium bumps are deposited, through which the two chips are welded together under controlled pressure and temperature, with an inter-chip spacing of 10 μm .

The performance of our sample is summarized in Table S1.

Wiring

Figure S2 illustrates the control wiring of our device. The control signals are generated at room temperature. Here, high-frequency signals (XY driving pulses 5.2 $\sim$ 5.6 GHz and readout input pulses 6.2 $\sim$ 6.7 GHz) are each generated by a pair of digital-to-analog Converter (DAC) channels connected to an IQ mixer. Meanwhile, low-frequency signals (qubit bias and coupler tuning) are generated directly by DAC channels. All these signals are transmitted through coaxial cables to the superconducting chip inside the dilution refrigerator. To reduce thermal noise from higher temperature stages, attenuators and low-pass filters are placed on the signal input lines at various temperature levels. The XY and Z control line are combined together through a mixer before being sent to the quantum processor. For the readout lines, three circulators are directly attached to the sample's readout port, asymmetrically diminishing the downward propagation of microwave signals. After that, the readout signals are amplified by Josephson junction amplifiers (JPA) located on the 10mK stage, which is controlled by a continuous microwave source and a direct current (DC) channel, then low-noise high-electron-mobility transistor (HEMT) located on 4K stage and room-temperature amplifiers gradually. The measurement of readout signals is executed at room temperature. They are first down-converted with IQ mixers, and then demodulated by analog-to-digital converters (ADC).

STATE PREPARATION

Establishment of the gauge field

To enable photon hopping between neighboring sites, the couplers are dynamically modulated on nanosecond timescales with coupling strength $J_{\text{eff}}(t) = 2J_{ij}^0 \cos[(\tilde{\omega}'_i - \tilde{\omega}_j)t + \varphi_{ij}]$, where the oscillation frequency is determined according to the detuning between photon boxes. In the rotating frame, the effective Hamiltonian becomes

$$\hat{H}_{\text{eff}}/\hbar = \sum_i \delta_i \hat{\sigma}_i^\dagger \hat{\sigma}_i + \sum_{i,j} J_{ij}^0 (e^{i\varphi_{ij}} \hat{\sigma}_j^\dagger \hat{\sigma}_i + e^{-i\varphi_{ij}} \hat{\sigma}_i^\dagger \hat{\sigma}_j), \quad (\text{S3})$$

where the effective qubit frequencies $\delta_i \equiv \tilde{\omega}_i - \tilde{\omega}'_i$ originate from the detuning between $\tilde{\omega}'_i$ and the actual transition frequencies $\tilde{\omega}_i$. More precise details can be found in the reference (38). The photon hopping rate $J_{ij}^0/2\pi$ can reach up to 5 MHz. In this article, we write the Hamiltonian in the two-level basis because of the sufficient on-site repulsion among photons. This is guaranteed by the large anharmonicities of the Plasmoniums ($\eta \sim -400$ MHz, see Table S1) with respect to the coupling strength. Also, we carefully arranged the qubit frequencies so that the oscillation frequencies are at least 50 MHz away from the transition to double-photon occupation ($|11\rangle \leftrightarrow |02\rangle$).

In Eq. S3, the photon hopping is regulated by a tunable phase φ_{ij} . When a photon runs over a closed loop, it accumulates a geometric phase equal to the sum of all the acquired phases within the loop. Analogous to electrons moving in a magnetic field, this can be interpreted as the effective magnetic field threading through the loop, in units of the reduced flux quanta. Moreover, we can manipulate the potential landscape of the lattice by modulating the oscillation frequency. A potential bump can be created by setting $\tilde{\omega}'_i$ to be lower than $\tilde{\omega}_i$, while a potential dip can be created by higher $\tilde{\omega}'_i$.

Local potential assisted adiabatic protocol

Here we delve into the details of the adiabatic preparation process, including the introduction of initial local potentials, and the choice of the adiabatic pulse shape.

The initial state of the adiabatic process is two photons localized in the red sites in Fig. S3A (site{2,2} and site{3,3}). At this time, the potential of the two sites is detuned V_0 , all couplings between lattice sites are turned off. The subsequent adiabatic process can be divided into two steps. The first is from $t = 0$ to $t = T_J$, during which we remain the local potential unchanged and gradually increase the hopping rate from 0 to 5 MHz. The second step is from $t = T_J$ to $t = T_{\text{End}}$, during which the hopping rate is fixed at 5 MHz, while the local potential gradually diminishes from V_0 to 0.

We first discuss the choice of local potential V_0 . Taking the magnetic flux at $\phi = 0.3$ for example, we illustrate the evolution of the system's energy gap (the energy difference between the first excited state and the ground state) throughout the entire adiabatic process (Fig. S3A). In absence of the local potential ($V_0 = 0$), the initial energy gap is 0, which doesn't support an adiabatic process. As the local potential is elevated, the initial energy gap widens gradually. At $V_0 = -1$ MHz, the energy gap starts from 1 MHz, then decreases to ~ 0.5 MHz, followed by an

increasement to ~ 0.7 MHz at $t = T_{\text{End}}$. Such a small energy gap is still not conducive to the adiabatic preparation of FQH states. However, when a large amount of local potential is introduced ($V_0 = -5$ MHz, $V_0 = -15$ MHz), the energy gap can be significantly increased. Through numerical simulation, we can estimate the impact of the value of local potentials on the fidelity of the final states, and thereby choose an appropriate V_0 as we will discuss later.

Next, we discuss the design of the adiabatic pulse. In the first step, the increase rate of hopping rate J is proportional to the square of the energy gap $\Delta(J)$

$$\frac{dJ}{dt} = \frac{\int_0^{5\text{MHz}} \frac{dJ}{\Delta^2(J)}}{T_J} \cdot \Delta^2(J), \quad (\text{S4})$$

Likewise, in the second step, the flattening rate of local potential V is determined as

$$\frac{dV}{dt} = \frac{\int_{V_0}^0 \frac{dV}{\Delta^2(V)}}{T_{\text{End}} - T_J} \cdot \Delta^2(V), \quad (\text{S5})$$

In order to obtain the complete information of the adiabatic pulse, we also need to determine the proportion of time utilized for the above two steps. In Fig. S3B, we numerically simulated the time evolution of the system and obtained the adiabatic fidelity (the fidelity between the final state and the target FQH state calculated with exact diagonalization) with respect to V_0 and T_J . Here, we fixed the total preparation time to $T_{\text{End}} = 1 \mu\text{s}$, which is limited by the coherence time of our system. We can obtain a robust result, demonstrating the capability to reach an adiabatic fidelity $> 95\%$ for the FQH state across a wide range of V_0 and T_J . As indicated by the red dot in Fig. S3B, we chose $V_0 = -15$ MHz for our experiment. The configuration of the adiabatic pulse utilized in experiment is depicted in Fig. S3C.

Post-selection

To attain a high fidelity of the prepared state, we implemented a two-step post-selection protocol in this experiment. Firstly, the residual photons at thermal equilibrium are suppressed by an active measurement of the qubits' initial state (62). The measurement module is placed before the injection of photons. Experimental results are only kept if all of the photon boxes are measured to be empty. After the measurement, we wait $1 \mu\text{s}$ for the readout resonator to ring down before photon injection. Secondly, at the end of the quantum evolution, the final photon number is extracted at the same time with the simultaneous quantum state measurement. We post-selected the results with correct number of photons remaining in the system. In this way, we are able to exclude most of the events affected by photon dissipation and readout error. This protocol is applied in the measurement of correlation values, quasiparticle generation and Hall conductivity. However, in the measurement of butterfly spectrum and density current, part of the information about photon occupation is missing. Instead, we implemented a measurement result correction based on the inverse positive operator-valued measure (POVM) matrices (63). It does not suppress the photon dissipation error but enhance the readout fidelity.

DATA MEASUREMENTS

Single-photon deflection in artificial magnetic field

Initially, a single photon is injected to the central two sites on the bottom row of the lattice, with the initial state configured as $(|01\rangle + |10\rangle)/\sqrt{2}$ for these two sites. After that, we activate the coupling directly to 5 MHz to synthesis the magnetic field. For each magnetic flux value, we monitored the photon density distribution across all 16 sites throughout the system's evolution. We calculated the centroid of the photon density and plotted its trajectory in Fig. 2A of the main text. The duration of evolution captured in this illustration is 65 ns. The detailed photon distribution during evolution can be found in the Supplementary Movie S1.

Butterfly spectrum

To estimate the single-photon energy spectrum, we initialize one site i in the lattice in a photon superposition state $(|0\rangle_i + |1\rangle_i)/\sqrt{2}$ and turn on the couplings between the sites. We then measure the observables $\chi_i = \langle \hat{\sigma}_i^X \rangle + i\langle \hat{\sigma}_i^Y \rangle$ after different durations of evolution. The full energy spectrum is extracted by performing a Fourier transformation of χ_i for each index i and average over all of them.

Fidelity measurement

As the exact tomography of the final states takes exponential time in experiment, here we roughly extracted the state fidelity through an inverted preparation process, as shown in Fig. S4. After the adiabatic preparation, we subjected the final state to an exact reversed adiabatic path, and measured the probability of the photons jointly occupying the initial sites, denoted as F . We use $f \sim \sqrt{F}$ to estimate the FQH state fidelity, which can reflect the impacts of decoherence and non-adiabatic processes. As shown in Fig. S4B, we numerically simulated f and $\sqrt{F}$ under different decoherence times and the duration of adiabatic processes. The results show that the difference between them is at most 2%.

Density correlation

The photon density correlation function is defined as the average spatial correlation between two photons as a function of their relative position $\mathbf{d}$, given by $g^{(2)}(\mathbf{d}) = \frac{N}{N-1} \mu_{\mathbf{i}, \mathbf{i}+\mathbf{d}} \frac{\langle \hat{n}_{\mathbf{i}} \hat{n}_{\mathbf{i}+\mathbf{d}} \rangle}{\langle \hat{n}_{\mathbf{i}} \rangle \langle \hat{n}_{\mathbf{i}+\mathbf{d}} \rangle}$, where $\hat{n}_{\mathbf{i}}$ is the photon number on site $\mathbf{i}$, and N is the total photon number. The average, denoted as μ , spans across all sites $\mathbf{i}$ with at least one site of $\mathbf{i}$ and $\mathbf{i} + \mathbf{d}$ belongs to the four central bulk sites of the system. We direct measure the photon density distributions to estimate the density correlation.

Density current

The photon density current is defined as the expectation value of two-site correlated operators, given by

$$\dot{J}_{((x,y),(x',y'))} = iJ_{((x,y),(x',y'))} \langle \hat{\sigma}_{x,y}^\dagger \hat{\sigma}_{x',y'} \rangle + \text{H.c.}, \quad (\text{S6})$$

where (x, y) and (x', y') represent the indices of two adjacent sites. It is mathematically equivalent to the velocity of photon flow between two isolated sites, which is governed by the following Hamiltonian

$$\hat{H}_{2q} = J_{((x,y),(x',y'))} (\hat{\sigma}_{x,y}^\dagger \hat{\sigma}_{x',y'} + \hat{\sigma}_{x,y} \hat{\sigma}_{x',y'}^\dagger). \quad (\text{S7})$$

As shown in Fig. S5A, we achieved isolation of the measured sites by flux-biasing their connecting counterparts to approximately 1 GHz off-resonance after the establishment of the ground state. The isolation occurs swiftly within 1 ns (limited by the rising edge of the square wave), during which time the measured sites remain stable. Subsequently, we measured the photon population of the isolated sites during 200 ns and fitted them with sinusoidal functions. The final results are calculated as the gradients of the fitted curve at the onset of isolation. A representative dataset for the current measurement is illustrated in Fig. S5B.

Hall conductivity

The quantum Hall conductivity σ_H is defined as the variation of photon bulk density ρ_{bulk} with respect to the magnetic flux density using the Středa's formula $\sigma_H = \sigma_0 \cdot \frac{\partial \rho_{\text{bulk}}}{\partial (\varphi/2\pi)}$, where $\sigma_0 = 2e/h$. We measure the photon density distributions at different magnetic flux to extract the value of Hall conductivity.

NUMERICAL SIMULATIONS

The numerical simulations in this work can be classified into three categories: exact diagonalization, pure state evolution under the system Hamiltonian H , and mixed state evolution under the Lindblad master equation with dephasing noise. The fidelities of the prepared states (gray solid line in Fig. 3D and Fig. S6B), the changes of photon population (gray dashed line in Fig. 5B) and the response of photon bulk density (gray dashed line in Fig. 5C) represent the numerical simulation results under the Lindblad master equation, with an average dephasing time of $T_2 = 6.2 \mu\text{s}$ of each site. Note that the experimentally measured average dephasing time at the idle point is $5.4 \mu\text{s}$, and we elevated it by 15% in numerical simulations to match the experimentally obtained fidelity. The photon deflection (solid line in Fig. 2A) and the butterfly spectrum (pseudocolor in Fig. 2B) represent the numerical simulation results of pure state evolution. The remaining numerical simulation results were obtained through exact diagonalization.

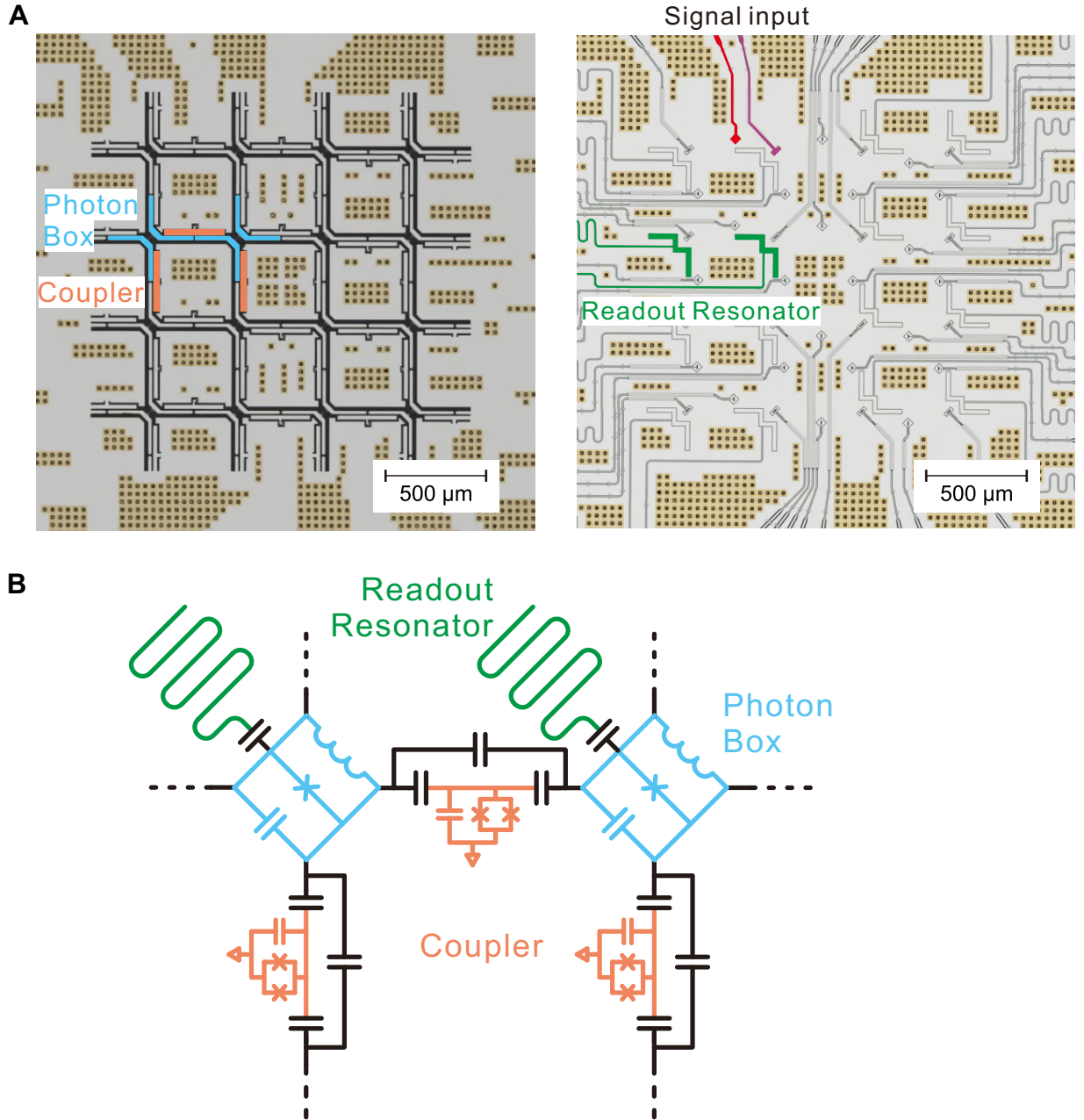


Fig. S1: (A) Optical micrograph of the sample. On the top chip, photon boxes (blue) are arranged in a two-dimensional lattice, which are coupled through Transmon-based couplers (orange). Signals are input from the bottom chip (red for couplers and purple for photon boxes). Dispersive measurement is enabled with the readout resonators (green). (B) Schematic of the circuit QED lattice, which is corresponding to the micrograph.

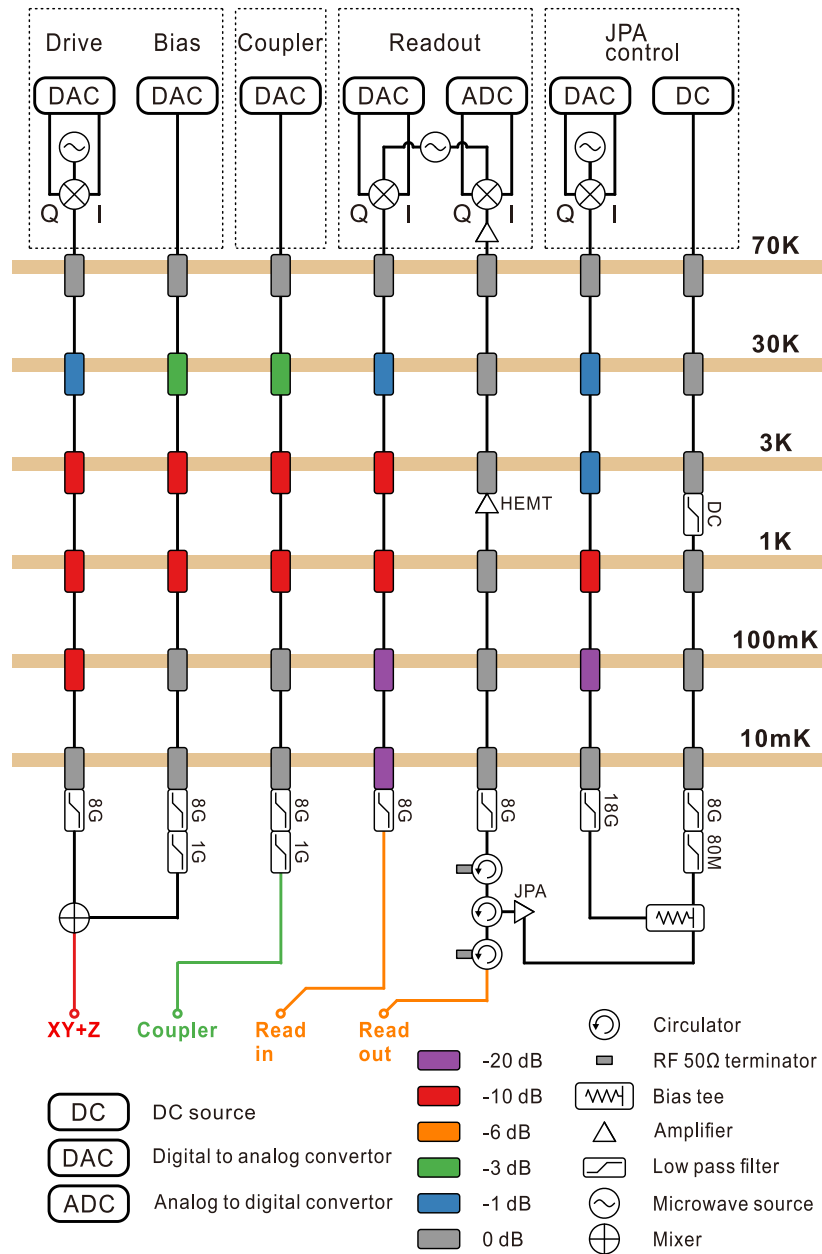


Fig. S2: Overview of the electronic devices and cryogenic wiring of our system.

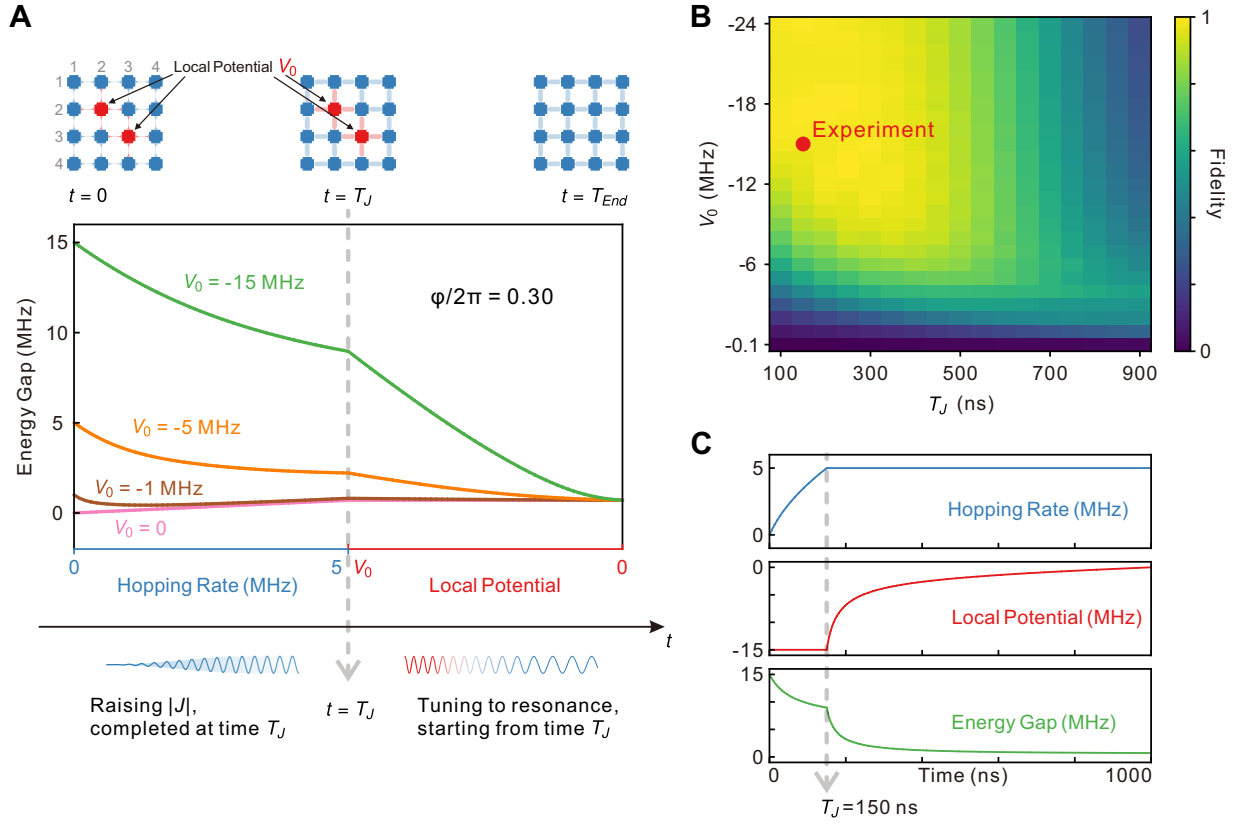


Fig. S3: Local potential assisted adiabatic protocol. (A) Evolution of the system's energy gap throughout the entire adiabatic process for different local potential choices. (B) The fidelity of preparing FQH states in numerical simulations with the different choices of local potential V_0 and the time required to increase the hopping rate T_J . The red dot marks the values used in the experiment. (C) The shape of the adiabatic ramp used in the experiment.

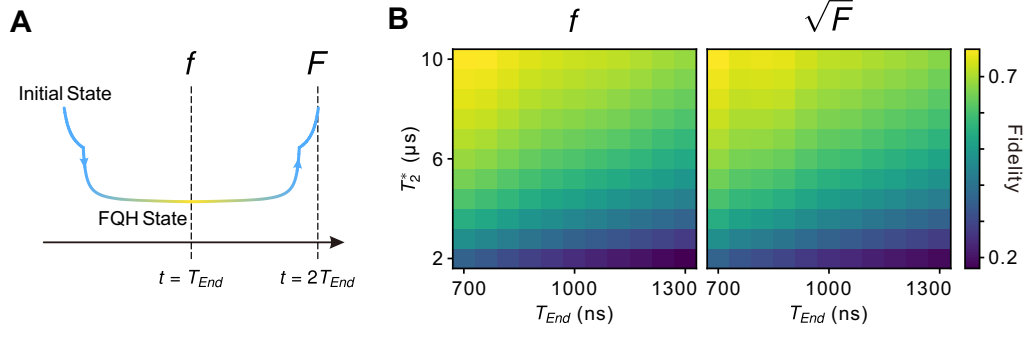


Fig. S4: Estimating the FQH state fidelity through an inverted preparation process. (A) Illustration of the inverted preparation. f denotes the prepared FQH state fidelity. F denotes the fidelity of the inverted prepared state, corresponding to the initial two-photon Fock state. (B) Numerical simulation results (magnetic flux $\phi = 0.3$) comparing f and $\sqrt{F}$ under different dephasing times T_2^* and adiabatic process durations T_{End} .

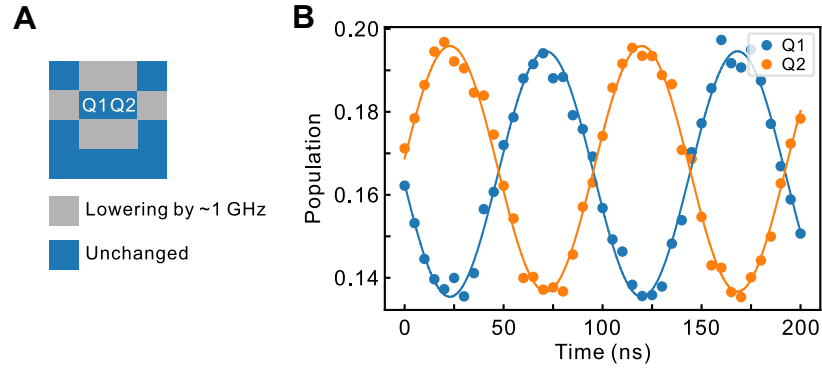


Fig. S5: Current measurement between qubit pairs. (A) Illustration of the isolation protocol. (B) Exchanging patterns of photons between the two isolated sites. Experimental data and fitting results are represented by dots and lines, respectively.

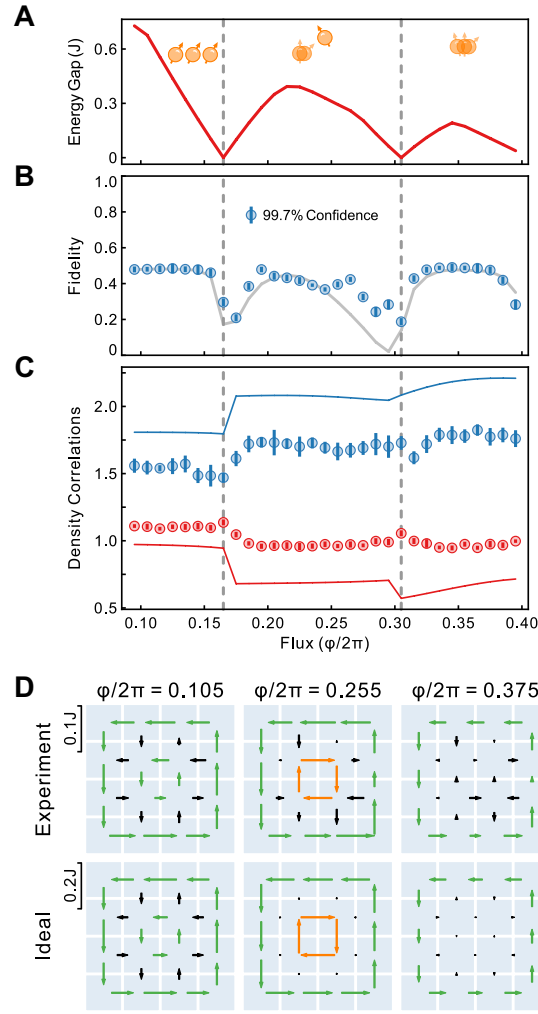


Fig. S6: Preparation and characterization of FQH states of $N=3$ photons. (A) Energy gaps as a function of magnetic flux. The gaps have two closing points, separating the phase into three parts: normal, crossover, and 3-photon FQH regime. In the new crossover regime, the lowest Landau level is likely to host two photons, similar to the 2-photon FQH state, while leaving the other photon in the higher level. (B) Ground state fidelities are measured with respect to the magnetic flux. Two breakdowns in fidelity present at the phase transition points. The gray solid line represents the simulation result with decoherence. (C) The density correlations of small and large distances at varying magnetic flux. When transition from normal to FQH states, the density correlations are suppressed at small distances while being enhanced at larger distances. The solid lines represent the exact values obtained from diagonalization. (D) The chiral density currents of the normal, crossover, and 3-photon FQH state, respectively. The crossover state shows a similar feature with the 2-photon FQH state (Fig. 4F). At 3-photon FQH state, the bulk currents and the radial currents flow between the boundary and the bulk are significantly suppressed.

	Q_{11}	Q_{12}	Q_{13}	Q_{14}	Q_{21}	Q_{22}	Q_{23}	Q_{24}	Q_{31}	Q_{32}	Q_{33}	Q_{34}	Q_{41}	Q_{42}	Q_{43}	Q_{44}
$\omega_{01}/2\pi$ (GHz)	5.352	5.524	5.279	5.576	5.452	5.272	5.423	5.286	5.377	5.561	5.260	5.531	5.517	5.321	5.508	5.427
$\eta/2\pi$ (MHz)	-461	-400	-493	-439	-418	-463	-413	-445	-475	-420	-412	-413	-460	-414	-426	-436
$\omega_r/2\pi$ (GHz)	6.201	6.317	6.434	6.552	6.244	6.350	6.487	6.596	6.216	6.352	6.457	6.578	6.273	6.398	6.526	6.648
T_1 (μ s)	21	29	25	20	29	42	26	28	29	23	30	28	24	23	30	37
T_2^* (μ s)	4.6	8.9	3.7	5.1	7.7	3.9	6.6	3.2	3.8	3.2	3.4	4.7	5.1	6.0	6.5	10.3
F_{00} (%)	99.4	99.3	99.1	98.2	97.7	97.5	97.7	97.1	99.1	99.1	98.1	97.0	97.0	97.6	96.7	96.4
F_{11} (%)	96.2	95.8	96.0	95.8	94.6	95.3	95.9	94.7	96.4	95.6	95.3	95.4	95.1	94.0	93.7	92.6

Table S1: Experimental parameters of all 16 qubits. Q_{ij} represents the qubit at the i -th row and j -th column. ω_{01} is the idle frequency of qubit and η is the anharmonicity. T_1 , the energy relaxation time, and T_2^* , the dephasing time extracted from Ramsey experiment, are measured at idle frequency. F_{00} (F_{11}) is the probability of correctly readout of qubit state in $|0\rangle$ ($|1\rangle$) after being prepared in $|0\rangle$ ($|1\rangle$).

Movie S1: Animated images show how single photons deflect in an artificial magnetic field. These images display the evolution of photon positions across the entire lattice under varying magnetic fluxes. The bottom left figure summarizes the trajectories of the centroids.

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